

ISC_sensor_block

July 18, 2025

```
[13]: %matplotlib qt
import mne

#mne.viz.set_3d_backend('pyvistaqt')
from mne.coreg import Coregistration
from mne.io import read_info

import numpy as np
#%matplotlib qt
import matplotlib
#matplotlib.use('qt5agg') # Or any other backend you want to use

import matplotlib.pyplot as plt

import pandas as pd
import os
from os.path import join as pathjoin
from pathlib import Path

import sys

from mne.preprocessing import ICA, corrmap, create_ecg_epochs, create_eog_epochs

from time import time

from autoreject import AutoReject

import glob

import psutil
import gc
from time import time

mne.set_log_level('INFO')

import json
```

```

from mne.channels import read_dig_polhemus_isotrak # Función para leer
↳archivos .pos

import re
# from mne.minimum_norm import apply_inverse, make_inverse_operator
#este codigo lo dejo comentado para acostumbrarme a su suso

import brainiak
from brainiak.isc import isc

import statsmodels
from statsmodels.stats.multitest import multipletests

import pickle

```

```

[14]: try:
        # Si se ejecuta como SCRIPT .py: usar __file__
        sys.path.append(os.path.abspath(os.path.join(os.path.dirname(__file__), "..
↳")))
    except NameError:
        # Si se ejecuta como NOTEBOOK Jupyter: usar path relativo
        sys.path.append("..") # sube un nivel desde la carpeta actual del notebook

# --- Configuración dinámica de rutas ---
from get_paths_MOUS import get_paths_MOUS

# Parámetros editables
disco = "g"
modality = "visual"
layer_script = "block"
subj = "sub-V1001"
#subj = sys.argv[1] ## name of participant list

# Generar variables automáticamente
path_dict = get_paths_MOUS(disco=disco, modality=modality,
↳layer_script=layer_script, subj=subj)
globals().update(path_dict)

# Mostrar todos los paths generados
print("\n Rutas generadas:")
for k, v in path_dict.items():
    print(f"{k:<20} → {v}")

```

Rutas generadas:

```
datadir          → g:\MOUS_204\MOUS_visual
general_datadir  → g:\MOUS_204
output_preproc   → g:\MOUS_204\MOUS_visual\output_preproc
mri_dir          → g:\MOUS_204\sub-V1001\anat
meg_dir          → g:\MOUS_204\sub-V1001\meg
preproc_path     → g:\MOUS_204\MOUS_visual\output_preproc\preproc_block
channels_structure_path → g:\MOUS_204\channels_structure
epochs_path      →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\epochs_block
ICA_path         →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\ICA_block
epochs_clean_path →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\epochs_clean_block
evoked_path      →
g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block
source_path      → g:\MOUS_204\MOUS_visual\output_source\source_block
raw_hsp_path     →
g:\MOUS_204\MOUS_visual\output_source\source_block\raw_hsp
fwd_path         → g:\MOUS_204\MOUS_visual\output_source\source_block\fwd
inverse_path     →
g:\MOUS_204\MOUS_visual\output_source\source_block\inverse
output_analysis  → g:\MOUS_204\MOUS_visual\output_analysis
analysis_path    → g:\MOUS_204\MOUS_visual\output_analysis\analysis_block
ACW_path        →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\acw_block
PLE_path        →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\PLE_block
ISC_path        →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\ISC_block
ISC_block_path   →
g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\ISC_block
```

```
[15]: #subjects = subj[:9]

subj = []

# Recorremos cada subdirectorio en la carpeta base
for subdirectorio in general_datadir.iterdir():
    # Comprobamos que el elemento sea un directorio y que su nombre comience
    ↪ con 'sub-A2'
    if modality == "visual":
        subject_prefix = 'sub-V1'
    elif modality == "auditory":
        subject_prefix = 'sub-A2'

    if subdirectorio.is_dir() and subdirectorio.name.startswith(subject_prefix):
```

```
# Añadimos el nombre del sujeto a la lista  
subj.append(subdirectorio.name)
```

```
print(subj)  
subjects= subj  
subjects
```

```
['sub-V1001', 'sub-V1002', 'sub-V1003', 'sub-V1004', 'sub-V1005', 'sub-V1006',  
'sub-V1007', 'sub-V1008', 'sub-V1009', 'sub-V1010', 'sub-V1011', 'sub-V1012',  
'sub-V1013', 'sub-V1015', 'sub-V1016', 'sub-V1017', 'sub-V1019', 'sub-V1020',  
'sub-V1022', 'sub-V1024', 'sub-V1025', 'sub-V1026', 'sub-V1027', 'sub-V1028',  
'sub-V1029', 'sub-V1030', 'sub-V1031', 'sub-V1032', 'sub-V1033', 'sub-V1034',  
'sub-V1035', 'sub-V1036', 'sub-V1037', 'sub-V1038', 'sub-V1039', 'sub-V1040',  
'sub-V1042', 'sub-V1044', 'sub-V1045', 'sub-V1046', 'sub-V1048', 'sub-V1049',  
'sub-V1050', 'sub-V1052', 'sub-V1053', 'sub-V1054', 'sub-V1055', 'sub-V1057',  
'sub-V1058', 'sub-V1059', 'sub-V1061', 'sub-V1062', 'sub-V1063', 'sub-V1064',  
'sub-V1065', 'sub-V1066', 'sub-V1068', 'sub-V1069', 'sub-V1070', 'sub-V1071',  
'sub-V1072', 'sub-V1073', 'sub-V1074', 'sub-V1075', 'sub-V1076', 'sub-V1077',  
'sub-V1078', 'sub-V1079', 'sub-V1080', 'sub-V1081', 'sub-V1083', 'sub-V1084',  
'sub-V1085', 'sub-V1086', 'sub-V1087', 'sub-V1088', 'sub-V1089', 'sub-V1090',  
'sub-V1092', 'sub-V1093', 'sub-V1094', 'sub-V1095', 'sub-V1097', 'sub-V1098',  
'sub-V1099', 'sub-V1100', 'sub-V1101', 'sub-V1102', 'sub-V1103', 'sub-V1104',  
'sub-V1105', 'sub-V1106', 'sub-V1107', 'sub-V1108', 'sub-V1109', 'sub-V1110',  
'sub-V1111', 'sub-V1113', 'sub-V1114', 'sub-V1115', 'sub-V1116', 'sub-V1117']
```

```
[15]: ['sub-V1001',  
       'sub-V1002',  
       'sub-V1003',  
       'sub-V1004',  
       'sub-V1005',  
       'sub-V1006',  
       'sub-V1007',  
       'sub-V1008',  
       'sub-V1009',  
       'sub-V1010',  
       'sub-V1011',  
       'sub-V1012',  
       'sub-V1013',  
       'sub-V1015',  
       'sub-V1016',  
       'sub-V1017',  
       'sub-V1019',  
       'sub-V1020',  
       'sub-V1022',  
       'sub-V1024',  
       'sub-V1025',  
       'sub-V1026',
```

'sub-V1027',
'sub-V1028',
'sub-V1029',
'sub-V1030',
'sub-V1031',
'sub-V1032',
'sub-V1033',
'sub-V1034',
'sub-V1035',
'sub-V1036',
'sub-V1037',
'sub-V1038',
'sub-V1039',
'sub-V1040',
'sub-V1042',
'sub-V1044',
'sub-V1045',
'sub-V1046',
'sub-V1048',
'sub-V1049',
'sub-V1050',
'sub-V1052',
'sub-V1053',
'sub-V1054',
'sub-V1055',
'sub-V1057',
'sub-V1058',
'sub-V1059',
'sub-V1061',
'sub-V1062',
'sub-V1063',
'sub-V1064',
'sub-V1065',
'sub-V1066',
'sub-V1068',
'sub-V1069',
'sub-V1070',
'sub-V1071',
'sub-V1072',
'sub-V1073',
'sub-V1074',
'sub-V1075',
'sub-V1076',
'sub-V1077',
'sub-V1078',
'sub-V1079',
'sub-V1080',

```

'sub-V1081',
'sub-V1083',
'sub-V1084',
'sub-V1085',
'sub-V1086',
'sub-V1087',
'sub-V1088',
'sub-V1089',
'sub-V1090',
'sub-V1092',
'sub-V1093',
'sub-V1094',
'sub-V1095',
'sub-V1097',
'sub-V1098',
'sub-V1099',
'sub-V1100',
'sub-V1101',
'sub-V1102',
'sub-V1103',
'sub-V1104',
'sub-V1105',
'sub-V1106',
'sub-V1107',
'sub-V1108',
'sub-V1109',
'sub-V1110',
'sub-V1111',
'sub-V1113',
'sub-V1114',
'sub-V1115',
'sub-V1116',
'sub-V1117']

```

```

[16]: def print5(*args):
        print(*(f"{x:.5f}" if isinstance(x, float) else x for x in args))

channels = pd.read_csv(channels_structure_path / f"channels_mag_{modality}.csv")
channels_mag=channels[channels[f"canal_efectivo_{modality}"].
↳notna()][f"canal_efectivo_{modality}"]

channels_mag=channels_mag.tolist()
print5(channels_mag)
indice_channels_efectivos = channels[channels[f"canal_efectivo_{modality}"].
↳notna()]["indice"]
indice_channels_efectivos=indice_channels_efectivos.tolist()
# del channels

```



```
'MRT27-4304', 'MRT31-4304', 'MRT32-4304', 'MRT33-4304', 'MRT34-4304',
'MRT35-4304', 'MRT36-4304', 'MRT37-4304', 'MRT41-4304', 'MRT42-4304',
'MRT43-4304', 'MRT44-4304', 'MRT45-4304', 'MRT46-4304', 'MRT47-4304',
'MRT51-4304', 'MRT52-4304', 'MRT53-4304', 'MRT54-4304', 'MRT55-4304',
'MRT56-4304', 'MRT57-4304', 'MZC01-4304', 'MZC02-4304', 'MZC03-4304',
'MZC04-4304', 'MZF01-4304', 'MZF02-4304', 'MZF03-4304', 'MZ001-4304',
'MZ002-4304', 'MZ003-4304', 'MZF01-4304']
```

```
[17]: valid_arrays = []
      valid_subjects = []

      n_subjects=len(subjects)
      # Condiciones
      for cond in ["zinnen", "woorden"]:

          valid_arrays = []
          valid_subjects = []

          for i in range(n_subjects):
              subj = subjects[i]
              file_path = evoked_path / f"{subj}_evoked_{cond}_block-ave.fif"

              try:
                  evokeds = mne.read_evokeds(file_path)
                  evoked = evokeds[0]
                  data_subj = evoked.pick("mag", exclude="bads").data # (n_dipoles, n_times)
                  data_subj_swapped = data_subj.T # (n_times, n_dipoles)

                  # Validar que los datos no sean basura
                  if np.all(data_subj_swapped == 0) or np.isnan(data_subj_swapped).all():
                      print(f"{subj} tiene solo ceros o NaNs, se omite.")
                      continue

                  valid_arrays.append(data_subj_swapped)
                  valid_subjects.append(subj)

              except FileNotFoundError:
                  print(f"Archivo no encontrado para {subj}: {file_path}")
                  continue
              except Exception as e:
                  print(f"Error al procesar {subj}: {e}")
                  continue

      array = np.stack(valid_arrays, axis=2) # (n_times, n_dipoles, n_subjects)
```

```

    print(f"Array ISC creado con forma {array.shape} ({len(valid_subjects)}  

    ↪ sujetos válidos)")

    iscs = isc(data=array, pairwise=False, summary_statistic=None,  

    ↪tolerate_nans=True)
    iscs_statistics=brainiak.isc.compute_summary_statistic(iscs,  

    ↪summary_statistic='mean', axis=0)

    ##bootstrap zinnen gives you the isc values, the confidence intervals, the  

    ↪p values and the distribution of the isc values
    ## so isc=iscs_bootstrap [0]

    iscs_bootstrap, ci,p, distribution= brainiak.isc.bootstrap_isc(iscs,  

    pairwise=False, summary_statistic='median', n_bootstraps=1000,  

    ci_percentile=95, side='right', random_state=None)

    print(f"iscs_statistics, {iscs_statistics}")
    print(f"iscs_bootstrap: {iscs}")
    print(f"ci: {ci}")
    print(f"p: {p}")
    print(f"distribution: {distribution}")
    print(f"iscs_statistics: {iscs_statistics}")

    threshold = 0.05

    # Encontrar los índices (canales) donde p[0] es menor que 0.05
    significant_channels = np.where(p < threshold)[0]

    # Contar cuántos canales cumplen la condición
    num_significant_channels = len(significant_channels)

    # Imprimir los canales significativos
    print(f"Canales significativos {cond} (p < 0.05):", significant_channels)
    print(f"Número total de canales significativos general en {cond}:",  

    ↪num_significant_channels)

    # Aplicar la corrección de Benjamini-Hochberg (FDR)
    _, p_adjusted, _, _ = multipletests(p, method='fdr_bh')
    significant_channels_adjusted = np.where(p_adjusted < threshold)[0]

    # Mostrar cuántos canales siguen siendo significativos
    num_significant_channels_adjusted = len(significant_channels_adjusted)
    print(f"Canales significativos {cond} después de FDR-BH:  

    ↪{significant_channels_adjusted} de {len(channels_mag)}")

```

```

    print(f"Número de Canales significativos {cond} después de FDR-BH:␣
↪{num_significant_channels_adjusted} de {len(channels_mag)}")
    dict = {
        f"iscs_cond_{cond}": iscs,
        f"iscs_statistics_{cond}": iscs_statistics,
        f"iscs_bootstrap_{cond}": iscs_bootstrap,
        f"ci_{cond}": ci,
        f"p_{cond}": p,
        f"distribution_{cond}": distribution,
        f"significant_channels_{cond}": significant_channels,
        f"num_significant_channels_{cond}": num_significant_channels,
        f"p_adjusted_{cond}": p_adjusted,
        f"significant_channels_adjusted_{cond}": significant_channels_adjusted,
        f"num_significant_channels_adjusted_{cond}":␣
↪num_significant_channels_adjusted
    }

    if cond=="zinnen":
        dict_zinnen = dict
    elif cond=="woorden":
        dict_woorden = dict

    # Definir la ruta de guardado
    pickle_path = ISC_path / f"dict_{cond}_{layer_script}.pkl"

    # Guardar el diccionario
    with open(pickle_path, 'wb') as f:
        pickle.dump(dict, f)

    print(f"Diccionario guardado en: {pickle_path}")

    dict={}

```

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1001_evoked_zinnen_block-ave.fif ...

Found the data of interest:

```

t =          0.00 ... 60000.00 ms (fix_zinnen)
0 CTF compensation matrices available
nave = 24 - aspect type = 100

```

No projector specified for this dataset. Please consider the method
self.add_proj.

No baseline correction applied

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1002_evoked_zinnen_block-ave.fif ...

Found the data of interest:

```

t =          0.00 ... 60000.00 ms (fix_zinnen)

```

```

    0 CTF compensation matrices available
    nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1003: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1003_evoked_zinnen_block-ave.fif
Archivo no encontrado para sub-V1004: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1004_evoked_zinnen_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1005_evoked_zinnen_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_zinnen)
        0 CTF compensation matrices available
        nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1006: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1006_evoked_zinnen_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1007_evoked_zinnen_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_zinnen)
        0 CTF compensation matrices available
        nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1008_evoked_zinnen_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_zinnen)
        0 CTF compensation matrices available
        nave = 23 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1009_evoked_zinnen_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_zinnen)
        0 CTF compensation matrices available
        nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-

```

```

V1010_evoked_zinnen_block-ave.fif ...
  Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_zinnen)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1011_evoked_zinnen_block-ave.fif ...
  Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_zinnen)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1012_evoked_zinnen_block-ave.fif ...
  Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_zinnen)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1013_evoked_zinnen_block-ave.fif ...
  Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_zinnen)
    0 CTF compensation matrices available
    nave = 23 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1015: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1015_evoked_zinnen_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1016_evoked_zinnen_block-ave.fif ...
  Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_zinnen)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1017: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1017_evoked_zinnen_block-ave.fif

```

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1019_evoked_zinnen_block-ave.fif ...

Found the data of interest:

t = 0.00 ... 60000.00 ms (fix_zinnen)

0 CTF compensation matrices available

nave = 23 - aspect type = 100

No projector specified for this dataset. Please consider the method
self.add_proj.

No baseline correction applied

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1020_evoked_zinnen_block-ave.fif ...

Found the data of interest:

t = 0.00 ... 60000.00 ms (fix_zinnen)

0 CTF compensation matrices available

nave = 24 - aspect type = 100

No projector specified for this dataset. Please consider the method
self.add_proj.

No baseline correction applied

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1022_evoked_zinnen_block-ave.fif ...

Found the data of interest:

t = 0.00 ... 60000.00 ms (fix_zinnen)

0 CTF compensation matrices available

nave = 24 - aspect type = 100

No projector specified for this dataset. Please consider the method
self.add_proj.

No baseline correction applied

Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1024_evoked_zinnen_block-ave.fif ...

Found the data of interest:

t = 0.00 ... 60000.00 ms (fix_zinnen)

0 CTF compensation matrices available

nave = 24 - aspect type = 100

No projector specified for this dataset. Please consider the method
self.add_proj.

No baseline correction applied

Archivo no encontrado para sub-V1025: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1025_evoked_zinnen_block-ave.fif

Archivo no encontrado para sub-V1026: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1026_evoked_zinnen_block-ave.fif

Archivo no encontrado para sub-V1027: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1027_evoked_zinnen_block-ave.fif

Archivo no encontrado para sub-V1028: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1028_evoked_zinnen_block-ave.fif

Archivo no encontrado para sub-V1029: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1029_evoked_zinnen_block-ave.fif

Archivo no encontrado para sub-V1030: g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-V1030_evoked_zinnen_block-ave.fif

[illegible]

Archivo no encontrado para sub-V1114: g:\MOUS_204\MOUS_visual\output_preproc\pre
 proc_block\evoked_block\sub-V1114_evoked_zinnen_block-ave.fif
 Archivo no encontrado para sub-V1115: g:\MOUS_204\MOUS_visual\output_preproc\pre
 proc_block\evoked_block\sub-V1115_evoked_zinnen_block-ave.fif
 Archivo no encontrado para sub-V1116: g:\MOUS_204\MOUS_visual\output_preproc\pre
 proc_block\evoked_block\sub-V1116_evoked_zinnen_block-ave.fif
 Archivo no encontrado para sub-V1117: g:\MOUS_204\MOUS_visual\output_preproc\pre
 proc_block\evoked_block\sub-V1117_evoked_zinnen_block-ave.fif
 Array ISC creado con forma (18001, 273, 15) (15 sujetos válidos)
 iscs_statistics, [2.13258647e-03 1.96933723e-03 -3.77194998e-03
 -4.56188401e-03
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 -1.34945862e-02 -5.16523782e-05 1.25471882e-02 1.60283450e-02
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2.89710290e-02 9.39060939e-02 3.96603397e-01 6.23376623e-01
9.99000999e-02 9.59040959e-02 9.12087912e-01 2.89710290e-02
3.92607393e-01 6.01398601e-01 1.00000000e+00 9.99000999e-04
5.99400599e-03 2.59740260e-02 6.13386613e-01 2.89710290e-02
2.69730270e-02 7.97202797e-01 7.94205794e-01 2.14785215e-01
9.29070929e-02 4.19580420e-01 2.69730270e-02 7.99200799e-03
9.99000999e-04 3.93606394e-01 3.93606394e-01 9.99000999e-04
1.99800200e-03 3.99600400e-03 9.29070929e-02 9.99000999e-04
1.03896104e-01 4.24575425e-01 1.06893107e-01 2.99700300e-03
3.82617383e-01 9.59040959e-02 3.94605395e-01 9.18081918e-01
9.99000999e-04 9.99000999e-04 9.99000999e-04 9.29070929e-01
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9.99000999e-04 2.20779221e-01 7.90209790e-01 9.74025974e-01
7.60239760e-01 4.02597403e-01 9.72027972e-01 1.00000000e+00
1.00000000e+00 1.00000000e+00 7.98201798e-01 1.12887113e-01
8.99100899e-01 8.79120879e-02 9.39060939e-02 2.15784216e-01
9.49050949e-02]
distribution: [[-0.0093718 -0.00880346 0.00117526 ... 0.00269225 -0.00235382
0.00878553]
[-0.01203429 -0.00054405 -0.01562467 ... 0.00287152 0.01442821
0.00499319]
[-0.01788471 -0.00660323 0.00117526 ... 0.00816829 -0.00235653
0.00499319]
...
[-0.00601995 -0.00880346 0.00045911 ... -0.00305653 -0.00235382
0.02765644]
[-0.00601995 -0.00660323 0.00871023 ... 0.00287152 0.0052658
-0.00722299]
[-0.01703378 -0.00880346 0.00259953 ... 0.00148625 -0.00015232
0.00878553]]
iscs_statistics: [ 2.13258647e-03 1.96933723e-03 -3.77194998e-03
-4.56188401e-03
-3.06567945e-04 7.11437646e-03 8.79071731e-03 -1.22818397e-02

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-1.25675479e-02	-1.19648165e-02	3.80268602e-03	4.89446878e-03
-3.58213055e-03	-1.28096757e-02	-9.12143333e-03	-2.23034175e-02
-9.57254038e-03	-2.32561754e-03	-8.55092718e-03	-4.87146564e-03
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8.84028167e-03	1.33605129e-02	1.38047648e-02	1.08766833e-02
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6.69136920e-03	-3.19024282e-03	-1.11592697e-02	-3.55343924e-03
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8.08128587e-03	1.18087727e-02	1.38814552e-02	1.02977625e-02
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-1.11798160e-02	-2.49842516e-02	-1.02652899e-02	1.14816219e-02
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-7.38532299e-04	3.35159577e-03	1.01207335e-02	2.25641905e-02
-2.36791042e-04	3.47926687e-03	8.40218729e-03	1.40799938e-02
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1.06963403e-02	4.62986141e-03	-7.51756200e-04	5.82708530e-03
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6.60152721e-03	4.44705446e-04	-1.85948849e-02	4.90629428e-03

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8.41561933e-03 6.93148718e-03 -2.51867548e-03 3.86401715e-03
6.46354795e-03 -7.02700391e-03 -6.35217573e-03 3.74350369e-03
-1.20712213e-03 5.01561619e-03 1.13657052e-02 8.82673798e-03
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2.05915944e-02 9.34832002e-03 -1.63857841e-02 -2.25242054e-02
-1.54806932e-02 7.29178246e-03 3.89267441e-02 3.32267541e-02
1.72652552e-02 -3.27941855e-03 -2.67827197e-02 -2.97334310e-02
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-8.17362368e-03 -1.74062394e-02 -1.27238423e-02 1.58141444e-02
1.75794269e-02 1.76751208e-02 -6.60789930e-04 -9.10818253e-03
-5.60781753e-03 7.35265516e-04 -5.49801844e-03 -1.32434246e-02
-2.38082397e-02 -1.65009667e-02 -1.70542410e-02 -1.61489993e-03
-3.90816310e-03 1.22738210e-02 9.93532867e-04 4.19395655e-03
8.73550400e-03]
Canales significativos zinnen (p < 0.05): [ 10 13 15 19 29 31 32 35 36
37 40 41 45 46 56 60 72 76
78 79 82 83 84 86 87 88 89 94 97 99 103 105 106 111 112 113
117 118 123 135 136 139 140 141 142 143 144 145 148 155 156 167 171 173
178 186 188 195 199 200 201 203 204 210 211 212 215 216 217 219 223 228
229 230 234 235 241 242 243 244 248 249 250 251 255 256]
Número total de canales significativos general en zinnen: 86
Canales significativos zinnen después de FDR-BH: [ 10 13 29 31 32 36 37
40 41 45 46 56 60 72 79 82 83 84
86 87 88 89 94 105 106 112 117 118 123 135 136 140 141 143 144 145
155 167 173 199 200 211 212 215 216 217 219 223 228 229 230 234 235 242
243 244 250 251 256] de 273
Número de Canales significativos zinnen después de FDR-BH: 59 de 273
Diccionario guardado en: g:\MOUS_204\MOUS_visual\output_analysis\analysis_block\
ISC_block\dict_zinnen_block.pkl
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1001_evoked_woorden_block-ave.fif ...
Found the data of interest:
t = 0.00 ... 60000.00 ms (fix_woorden)
0 CTF compensation matrices available
nave = 20 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1002_evoked_woorden_block-ave.fif ...
Found the data of interest:
t = 0.00 ... 60000.00 ms (fix_woorden)
0 CTF compensation matrices available

```

```

    nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1003: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1003_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1004: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1004_evoked_woorden_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1005_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ...   60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1006: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1006_evoked_woorden_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1007_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ...   60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1008_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ...   60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1009_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ...   60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 21 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1010_evoked_woorden_block-ave.fif ...

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```

Found the data of interest:
    t =          0.00 ... 60000.00 ms (fix_woorden)
    0 CTF compensation matrices available
    nave = 21 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1011_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 23 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1012_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1013_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 23 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1015: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1015_evoked_woorden_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1016_evoked_woorden_block-ave.fif ...
    Found the data of interest:
        t =          0.00 ... 60000.00 ms (fix_woorden)
        0 CTF compensation matrices available
        nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1017: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1017_evoked_woorden_block-ave.fif
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-

```

```

V1019_evoked_woorden_block-ave.fif ...
  Found the data of interest:
    t =      0.00 ...  60000.00 ms (fix_woorden)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1020_evoked_woorden_block-ave.fif ...
  Found the data of interest:
    t =      0.00 ...  60000.00 ms (fix_woorden)
    0 CTF compensation matrices available
    nave = 20 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1022_evoked_woorden_block-ave.fif ...
  Found the data of interest:
    t =      0.00 ...  60000.00 ms (fix_woorden)
    0 CTF compensation matrices available
    nave = 22 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Reading g:\MOUS_204\MOUS_visual\output_preproc\preproc_block\evoked_block\sub-
V1024_evoked_woorden_block-ave.fif ...
  Found the data of interest:
    t =      0.00 ...  60000.00 ms (fix_woorden)
    0 CTF compensation matrices available
    nave = 24 - aspect type = 100
No projector specified for this dataset. Please consider the method
self.add_proj.
No baseline correction applied
Archivo no encontrado para sub-V1025: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1025_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1026: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1026_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1027: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1027_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1028: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1028_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1029: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1029_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1030: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1030_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1031: g:\MOUS_204\MOUS_visual\output_preproc\pre

```



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proc_block\evoked_block\sub-V1114_evoked_woorden_block-ave.fif
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proc_block\evoked_block\sub-V1115_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1116: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1116_evoked_woorden_block-ave.fif
Archivo no encontrado para sub-V1117: g:\MOUS_204\MOUS_visual\output_preproc\pre
proc_block\evoked_block\sub-V1117_evoked_woorden_block-ave.fif
Array ISC creado con forma (18001, 273, 15) (15 sujetos válidos)
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  5.25244459e-03  6.71443640e-03  1.40641591e-02  1.65650043e-02
  1.18721359e-02  1.83188714e-02  9.59073869e-03  1.87953345e-02
  1.06711282e-02  8.92630977e-03  1.52536313e-02  1.41363265e-02
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  2.89887883e-02  2.48731823e-02  1.13066601e-02  1.32807895e-02
  1.39969096e-02  2.38116893e-02  2.83549505e-02  2.17271788e-02
  1.60229644e-02 -2.46425407e-03 -5.33512123e-03 -6.96906151e-03
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77 78 81 82 83 84 86 87 88 89 90 92 93 94 96 112 118 121
122 125 130 132 133 134 140 141 143 144 145 150 168 169 180 184 186 192
198 201 202 204 207 210 213 215 220 221 224 237 239 240 252 254 256 257
258 263 269 270]
Número total de canales significativos general en palabras: 76
Canales significativos palabras después de FDR-BH: [ 1 14 23 46 52 56 58
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88 89 90 92 93 94 112 118 122 133 140 141 144 145 150 180 184 192
201 210 213 215 220 221 224 237 240 252 254 256 269 270] de 273
Número de Canales significativos palabras después de FDR-BH: 50 de 273
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